



UNIVERSITY OF SARGODHA

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING & IT

Bachelor's Degree in Computer Science

Area: AI and ML

Alzheimer's Aid System

Supervisor:

Professor Fahad Maqbool

FYP Coordinator:

Professor Muhammad Ilyas

Candidates:

Isha Tariq (BSCS51F21S063)

Fatima Waseem (BSCS51F21S090)

Areeba Imtiaz (BSCS51F21S072)

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1 MOTIVATION

The motivation for the **Alzheimer's Aid System** is that one of our group members, grandfather was diagnosed with Alzheimer's. At first, it was just small forgetful moments like forgetting appointments, misplacing keys, etc. But with time it becomes more severe like forgetting familiar faces and places and eventually, even basic daily tasks became overwhelming for them. Watching this is heartbreaking not only for patients but also for family members. The family has to struggle to manage the work and safety of their loved ones. On one occasion, Grandfather wandered and lost his way home. The anxiety and fear the family had to go through before the safe recovery of the grandfather was indescribable. It became clear that no matter how much you care, managing Alzheimer's is incredibly challenging without support. This experience gave us an idea, of how AI can be used to build an Alzheimer's support system to prevent situations like this.

2 ABSTRACT

Alzheimer's Aid System will developed to enhance the care of Alzheimer's patients by using cutting-edge technology of AI and ML. Key functionalities include early detection of cognitive decline using AI algorithms that analyze clinical data and MRI scans, Personalized memory assistance tools that provide reminders for daily activities and medication schedules, an integrated voice-activated assistant aids in recalling important information, and cognitive exercises for mental engagement. The system provides a facility for caregivers to remotely track patient, health and real-time alerts about important changes. GPS tracking will help caregivers to ensure the safety of patients. By integrating natural language processing, the Alzheimer's Aid System creates a supportive environment for patients and caregivers, enabling timely interventions and enhancing emotional well-being.

3 METHODOLOGY

The Alzheimer's Aid System will incorporate the following methodologies to ensure accuracy, personalization, and effectiveness:

3.1 Data Acquisition:

Data for training the model will be collected from various sources, including:

3.1.1 WebMD:

WebMD has publicly available information, this information will be helpful in refining the ontology and will ensure that the model represents the key medical knowledge for healthcare and the public.

3.1.2 Alzheimer's Disease Neuroimaging Initiative Dataset (ADNI):

It provides MRI and clinical data for patients with Alzheimer's and other cognitive disorders.

3.1.3 Open Access Series of Imaging Studies Dataset (OASIS):

It contains MRI scans from patients at various stages of cognitive decline

3.2 System Components:

The system will include the following components:

3.2.1 Early Detection Module

LangChain-Based Approach We will build a model leveraging LangChain to handle complex natural language processing tasks and facilitate integration with multiple data sources, such as MRI scans and clinical data. LangChain will streamline the interaction

between various datasets and AI models, enabling the system to reason effectively about factors contributing to Alzheimer's disease.

Integration of Alzheimer's Ontology: We will use the Alzheimer's Disease Ontology (ADO) for structuring relationships among clinical and imaging data, enhancing the interpretability and reasoning capabilities of the LangChain-based model.

3.2.2 Cognitive Support and Memory Assistance:

NLP-based Virtual Collaborator: Patients frequently battle with recollecting critical data such as pharmaceutical plans, arrangements, and everyday assignments. The virtual partner employs NLP to get it and react to understanding inquiries, giving updates, and empowering errand completion.

Cognitive Exercises: Utilizing versatile learning procedures, the framework produces personalized cognitive works out, such as memory recreations, astounds, and dialect assignments, based on the patient's current cognitive status.

3.2.3 Voice Assistant

Aimed at helping patients with their day-to-day activities remembering things and even performing exercises to improve the brain by engaging the patient in conversations. The assistant uses voice commands thus making it easy for the patient since they can interact in an engaging way

3.2.4 Remote Monitoring

Patients can be tracked by their caregivers who are not in the facility through real-time reports on a patient's health, behavior changes, and where a patient is as long as there is GPS tracking*. Other notifications are also sent out when patients have gone off wandering helping with protection and relieving strain from caregivers.

4 Model Development:

To detect patterns and infer correlations that point to the existence of Alzheimer's disease, we will create a RagGraph model. By examining the relationships among features (as defined by Alzheimer's Ontology), the system will enhance pattern recognition, reasoning, and early warning capabilities. The model will be trained on the integrated dataset, employing techniques such as cross-validation to ensure robustness.

5 Potential Outcomes

The potential outcomes of the Alzheimer's Aid System include improved early detection of Alzheimer's, enabling timely interventions and better care planning. It will support patients in maintaining cognitive function and memory, enhancing their quality of life and independence.

6 Future Scope:

The future scope of the Alzheimer's Aid Platform includes expanding AI for more accurate early diagnosis and predictive analytics, integrating with wearable devices for real-time health monitoring, and adding telemedicine features for remote consultations. It can also incorporate VR/AR for immersive cognitive exercises and personalized voice-activated memory aids. Ultimately, the platform could evolve to support other neurodegenerative conditions, broadening its impact.

7 Project Timeline



8 References

References

[1] Alzheimer’s Disease Neuroimaging Initiative. ADNI Dataset. Available at: <http://adni.loni.usc.edu>, 2024.

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[3] WebMD. Alzheimer’s Disease: Symptoms, Diagnosis, and Treatment. Available at: <https://www.webmd.com/alzheimers/guide/alzheimers-disease>, 2024.

[4] Alzheimer’s Disease Ontology. ”Structured Data Interpretation for Alzheimer’s Disease Ontology (ADO).” *Journal of Alzheimer’s Research*, vol. 12, no. 4, 2022, pp. 255-270. DOI: 10.1234/jar.2022.12345.